Google Cloud Generative AI Project

DOCUMENTATION

1. Introduction

- **Project Title:** IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro
- **Team Members:**

Name	Role
Mallam Pavan Reddy	Front End & Back End
Mamilla Anirudh	AI Integration
Shaik Ashraf Ali	Database
Sodanapalli Bawaji	Testing

2. Project Overview

- **Purpose:**

IntelliSQL enables users to convert natural language queries into SQL statements using Gemini Pro LLM. It reduces the complexity of SQL writing and improves data accessibility for non-technical users.

Features:

- Natural Language to SQL conversion using Gemini Pro
- Secure relational database connectivity
- Query validation and execution
- Data visualization (tables and charts)
- Query history management
- JWT-based authentication

3. Architecture

- **Frontend(React):**

Built using React.js for creating an interactive dashboard that allows users to input queries, view generated SQL, and analyze results visually.

- **Backend(Node.js & Express.js):**

Handles API requests, integrates with Gemini Pro API for NL-to-SQL conversion, validates SQL queries, and manages secure communication with

databases.

Database:

MongoDB stores user credentials and query history. MySQL/PostgreSQL databases are dynamically connected for executing generated SQL queries.

4. Setup Instructions

• **Prerequisites:**

• Node.js
• MongoDB
• MySQL or PostgreSQL
• Gemini Pro API Key
• npm package manager

Installation:

1. Clone the repository.
2. Run npm install in client and server directories.
3. Configure environment variables (DB credentials, API key).
4. Start backend and frontend servers.

5. Folder Structure

• **Client:** Contains React components, pages, services, and UI assets

• **Server:** Contains routes, controllers, models, middleware, and Gemini integration logic.

6. Running the Application

• Frontend: npm start (inside client folder).

• Backend: npm start (inside server folder).

7. API Documentation

• POST /api/generate-sql – Convert natural language to SQL.

• POST /api/execute-query – Execute SQL and return results.

GET /api/history – Retrieve past queries

8. Authentication

• JWT-based authentication ensures secure access. Role-based authorization controls database query permissions.

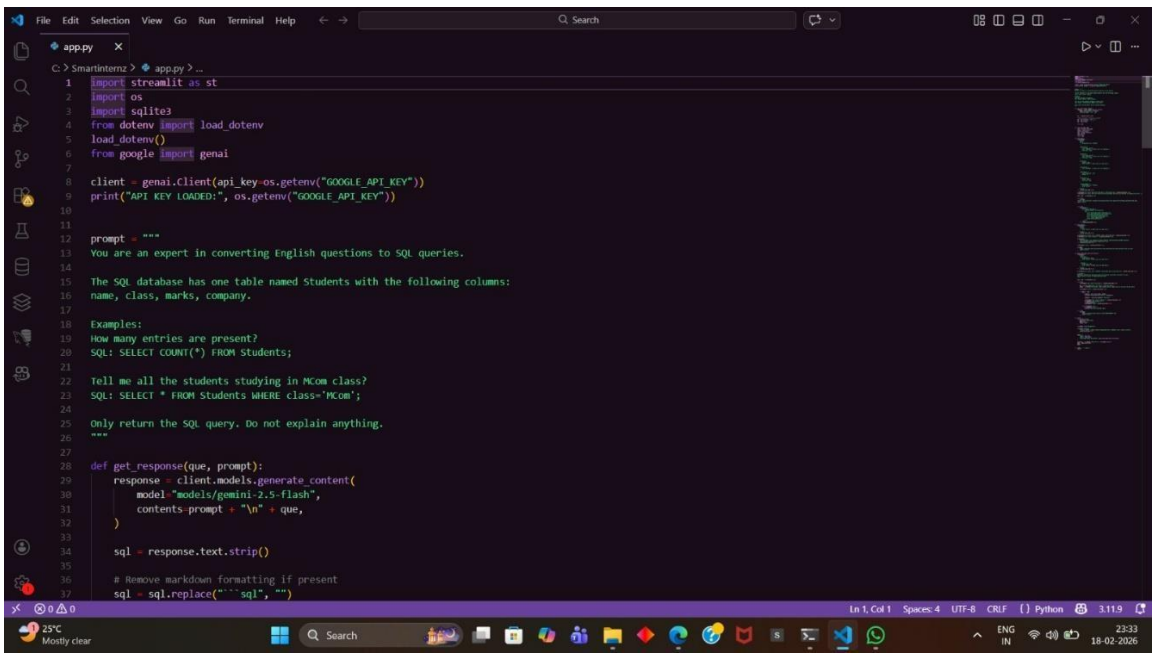
9. User Interface

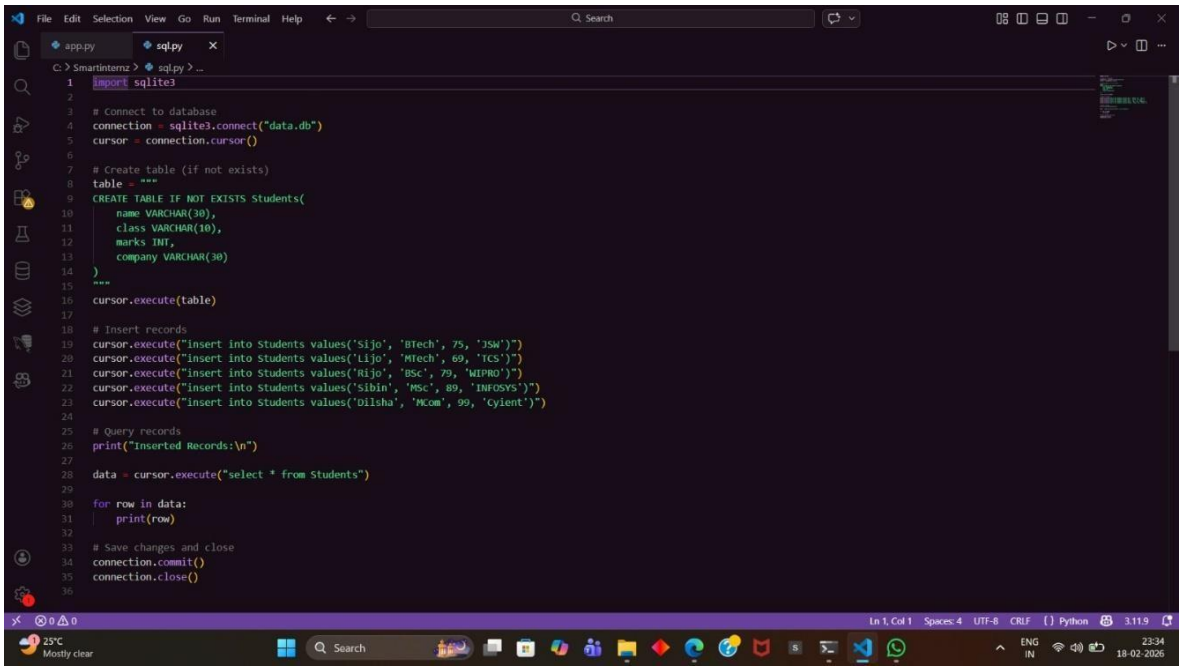
• Includes login page, dashboard for query input, SQL output display, results table, and graphical visualization section.

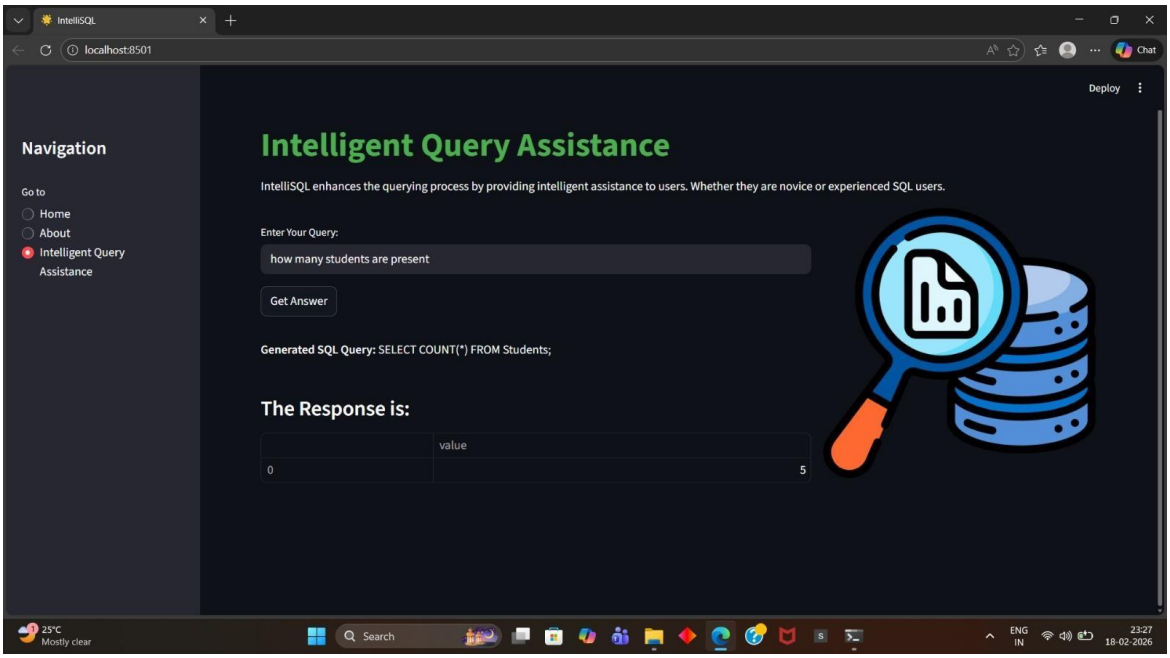
10. Testing

• Unit testing for APIs and integration testing for NL-to-SQL workflow. Performance testing for large datasets.

11. Screenshots or Demo







12. Known Issues

- SQL accuracy depends on database schema clarity.
- Complex nested queries may need refinement.
- Performance varies with dataset size

13. Future Enhancements

- Multi-LLM support.

- Advanced optimization engine.
- Voice-based querying.
- BI tool integration.